NATIONAL QUANTUM MISSION AND QUANTUM TECHNOLOGY

BIT AND QUBIT CARD → QUANTUM PRINCIPLE SPLIT → ALGORITHM MECHANISM RAIL → DECOHERENCE ENGINEERING STACK → PHYSICAL-TO-LOGICAL LADDER → FOUR TECHNOLOGY FAMILIES

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW · ● SUBORDINATE EXTRA · ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 BIT AND QUBIT CARD

BIT AND QUBIT CARD · CLASSICAL BIT · RECORDED AS 0 OR 1 · AMPLITUDES OVER BASIS STATES · ONE CLASSICAL OUTCOME WITH PROBABILITY · NOT THE DEFINING DIFFERENCE · STATE DESCRIPTION IS NOT TWO STORED CLASSICAL BITS

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
CLASSICAL BIT → recorded as 0 or 1	MEASUREMENT → one classical outcome with probability	RULE → state description is not two stored classical bits
QUBIT → amplitudes over basis states	SPEED → not the defining difference	CLASSICAL BIT → recorded as 0 or 1

SYSTEM / DEVICE / INSTITUTION
- CLASSICAL BIT → recorded as 0 or 1
- QUBIT → amplitudes over basis states

DECISIVE TECHNICAL DISTINCTION
- MEASUREMENT → one classical outcome with probability
- SPEED → not the defining difference

STATUS / UNIT / EVIDENCE LIMIT
- RULE → state description is not two stored classical bits
- CLASSICAL BIT → recorded as 0 or 1

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → state description is not two stored classical bits.

01

STAGE 01 QUANTUM PRINCIPLE SPLIT

QUANTUM PRINCIPLE SPLIT · ONE SYSTEM IN A COMBINATION OF BASIS STATES · JOINT STATE OF TWO OR MORE SYSTEMS · ARBITRARY UNKNOWN STATE CANNOT BE COPIED PERFECTLY · NO SIGNALLING · ENTANGLEMENT DOES NOT TRANSMIT MESSAGES FASTER · EACH PRINCIPLE ANSWERS A DIFFERENT QUESTION

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
SUPERPOSITION → one system in a combination of basis states	NO-CLONING → arbitrary unknown state cannot be copied perfectly	RULE → each principle answers a different question
ENTANGLEMENT → joint state of two or more systems	NO SIGNALLING → entanglement does not transmit messages faster	SUPERPOSITION → one system in a combination of basis states

SYSTEM / DEVICE / INSTITUTION
- SUPERPOSITION → one system in a combination of basis states
- ENTANGLEMENT → joint state of two or more systems

DECISIVE TECHNICAL DISTINCTION
- NO-CLONING → arbitrary unknown state cannot be copied perfectly
- NO SIGNALLING → entanglement does not transmit messages faster

STATUS / UNIT / EVIDENCE LIMIT
- RULE → each principle answers a different question
- SUPERPOSITION → one system in a combination of basis states

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → each principle answers a different question.

02

STAGE 02 ALGORITHM MECHANISM RAIL

ALGORITHM MECHANISM RAIL · PREPARE STATE · INITIALISE QUBITS · QUANTUM GATES · TRANSFORM AMPLITUDES · SUPPRESS AND REINFORCE PATHS · OBTAIN CLASSICAL SAMPLE · SUPERPOSITION DOES NOT REVEAL EVERY ANSWER

PREPARE STATE → initialise qubits → QUANTUM GATES → transform amplitudes → INTERFERENCE → suppress and reinforce paths → MEASUREMENT → obtain classical sample → TRAP → superposition does not reveal every answer

STARTING CONDITION / INPUT
- PREPARE STATE → initialise qubits
- QUANTUM GATES → transform amplitudes

SCIENTIFIC OR ENGINEERING MECHANISM
- INTERFERENCE → suppress and reinforce paths
- MEASUREMENT → obtain classical sample

OUTPUT / FAILURE BOUNDARY
- TRAP → superposition does not reveal every answer
- PREPARE STATE → initialise qubits

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → superposition does not reveal every answer.

03

STAGE 03 DECOHERENCE ENGINEERING STACK

DECOHERENCE ENGINEERING STACK · ENVIRONMENTAL INTERACTION · PHASE INFORMATION DEGRADES · COHERENCE TIME · FINITE COMPUTATION WINDOW · REDUCE DISTURBANCE · CONTROL ELECTRONICS · EXECUTE ACCURATE GATES

STARTING CONDITION / INPUT
- ENVIRONMENTAL INTERACTION → phase information degrades
- COHERENCE TIME → finite computation window

SCIENTIFIC OR ENGINEERING MECHANISM
- CRYOGENICS / VACUUM / SHIELDING → reduce disturbance
- CONTROL ELECTRONICS / MATERIALS → execute accurate gates

OUTPUT / FAILURE BOUNDARY
- OUTCOME → engineering, not theory alone, limits scale
- ENVIRONMENTAL INTERACTION → phase information degrades

- PREPARE STATE → initialise qubits
- QUANTUM GATES → transform amplitudes
- INTERFERENCE → suppress and reinforce paths
- MEASUREMENT → obtain classical sample
- TRAP → superposition does not reveal every answer
- PREPARE STATE → initialise qubits

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → superposition does not reveal every answer.

03

STAGE 03

DECOHERENCE ENGINEERING STACK

DECOHERENCE ENGINEERING STACK · ENVIRONMENTAL INTERACTION · PHASE INFORMATION DEGRADES · COHERENCE TIME · FINITE COMPUTATION WINDOW · REDUCE DISTURBANCE · CONTROL ELECTRONICS · EXECUTE ACCURATE GATES

STARTING CONDITION / INPUT

- ENVIRONMENTAL INTERACTION → phase information degrades
- COHERENCE TIME → finite computation window

SCIENTIFIC OR ENGINEERING MECHANISM

- CRYOGENICS / VACUUM / SHIELDING → reduce disturbance
- CONTROL ELECTRONICS / MATERIALS → execute accurate gates

OUTPUT / FAILURE BOUNDARY

- OUTCOME → engineering, not theory alone, limits scale
- ENVIRONMENTAL INTERACTION → phase information degrades

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → engineering, not theory alone, limits scale.

04

STAGE 04

PHYSICAL-TO-LOGICAL LADDER

PHYSICAL-TO-LOGICAL LADDER · PHYSICAL QUBIT · NOISY HARDWARE ELEMENT · ERROR CHARACTERISATION · MEASURE FAILURE CHANNELS · ERROR-CORRECTING CODE · REDUNDANT ENCODING · LOGICAL QUBIT

PHYSICAL QUBIT → noisy hardware element

ERROR CHARACTERISATION → measure failure channels → ERROR-CORRECTING CODE → redundant encoding → LOGICAL QUBIT → protected information unit

STARTING CONDITION / INPUT

- PHYSICAL QUBIT → noisy hardware element
- ERROR CHARACTERISATION → measure failure channels

SCIENTIFIC OR ENGINEERING MECHANISM

- ERROR-CORRECTING CODE → redundant encoding
- LOGICAL QUBIT → protected information unit

OUTPUT / FAILURE BOUNDARY

- RULE → raw qubit count is not useful capability
- PHYSICAL QUBIT → noisy hardware element

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → raw qubit count is not useful capability.

05

STAGE 05

FOUR TECHNOLOGY FAMILIES

FOUR TECHNOLOGY FAMILIES · SELECTED INFORMATION-PROCESSING ALGORITHMS · QUANTUM-STATE LINKS AND KEY DISTRIBUTION · PRECISION MEASUREMENT · ENABLING HARDWARE SUBSTRATE · QUANTUM TECHNOLOGY IS NOT COMPUTING ALONE

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
COMPUTING → selected information-processing algorithms	SENSING / METROLOGY → precision measurement	TRAP → quantum technology is not computing alone
COMMUNICATION → quantum-state links and key distribution	MATERIALS / DEVICES → enabling hardware substrate	COMPUTING → selected information-processing algorithms

SYSTEM / DEVICE / INSTITUTION

- COMPUTING → selected information-processing algorithms
- COMMUNICATION → quantum-state links and key distribution

DECISIVE TECHNICAL DISTINCTION

- SENSING / METROLOGY → precision measurement
- MATERIALS / DEVICES → enabling hardware substrate

STATUS / UNIT / EVIDENCE LIMIT

- TRAP → quantum technology is not computing alone
- COMPUTING → selected information-processing algorithms

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → quantum technology is not computing alone.

06

STAGE 06

QKD SECURITY CHAIN

QKD SECURITY CHAIN · QUANTUM STATES · DISTRIBUTE KEY MATERIAL · DISTURBANCE TEST · REVEAL POSSIBLE INTERCEPTION · AUTHENTICATED ENDPOINTS · PREVENT IMPERSONATION · CLASSICAL ENCRYPTION

QUANTUM STATES → distribute key material → DISTURBANCE TEST → reveal possible interception → AUTHENTICATED ENDPOINTS → prevent impersonation → CLASSICAL ENCRYPTION → protects message content → RULE → QKD alone is not a complete secure network

STARTING CONDITION / INPUT

- QUANTUM STATES → distribute key material
- DISTURBANCE TEST → reveal possible interception

SCIENTIFIC OR ENGINEERING MECHANISM

- AUTHENTICATED ENDPOINTS → prevent impersonation
- CLASSICAL ENCRYPTION → protects message content

OUTPUT / FAILURE BOUNDARY

- RULE → QKD alone is not a complete secure network
- QUANTUM STATES → distribute key material

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → QKD alone is not a complete secure network.

06

STAGE 06 QKD SECURITY CHAIN

QKD SECURITY CHAIN · QUANTUM STATES · DISTRIBUTE KEY MATERIAL · DISTURBANCE TEST · REVEAL POSSIBLE INTERCEPTION · AUTHENTICATED ENDPOINTS · PREVENT IMPERSONATION · CLASSICAL ENCRYPTION

QUANTUM STATES → distribute key material → DISTURBANCE TEST → reveal possible interception → AUTHENTICATED ENDPOINTS → prevent impersonation → CLASSICAL ENCRYPTION → protects message content → RULE → QKD alone is not a complete secure network

STARTING CONDITION / INPUT

- QUANTUM STATES → distribute key material
- DISTURBANCE TEST → reveal possible interception

SCIENTIFIC OR ENGINEERING MECHANISM

- AUTHENTICATED ENDPOINTS → prevent impersonation
- CLASSICAL ENCRYPTION → protects message content

OUTPUT / FAILURE BOUNDARY

- RULE → QKD alone is not a complete secure network
- QUANTUM STATES → distribute key material

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → QKD alone is not a complete secure network.

07

STAGE 07 QKD VERSUS PQC

QKD VERSUS PQC · QUANTUM HARDWARE AND PHYSICAL LINK · CLASSICAL QUANTUM-RESISTANT ALGORITHMS · QKD PURPOSE · SPECIALISED KEY DISTRIBUTION · PQC PURPOSE · BROAD STANDARDS AND SOFTWARE MIGRATION · COMPLEMENTARY TOOLS WITH DIFFERENT TIMELINES

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
QKD → quantum hardware and physical link	QKD PURPOSE → specialised key distribution	POLICY → complementary tools with different timelines
PQC → classical quantum-resistant algorithms	PQC PURPOSE → broad standards and software migration	QKD → quantum hardware and physical link

SYSTEM / DEVICE / INSTITUTION

- QKD → quantum hardware and physical link
- PQC → classical quantum-resistant algorithms

DECISIVE TECHNICAL DISTINCTION

- QKD PURPOSE → specialised key distribution
- PQC PURPOSE → broad standards and software migration

STATUS / UNIT / EVIDENCE LIMIT

- POLICY → complementary tools with different timelines
- QKD → quantum hardware and physical link

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PQC PURPOSE → broad standards and software migration.

08

STAGE 08 NQM INSTITUTION MAP

NQM INSTITUTION MAP · MINISTRY OF SCIENCE AND TECHNOLOGY · MINISTERIAL UMBRELLA · IMPLEMENTING DEPARTMENT · NATIONAL CAPABILITY MISSION · FOUR VERTICAL ECOSYSTEM ANCHORS · DEPARTMENT, MISSION AND HUB ARE DISTINCT

MINISTRY OF SCIENCE AND TECHNOLOGY → ministerial umbrella

DST → implementing department · NQM → national capability mission → T-HUBS → four vertical ecosystem anchors

CONCEPT / ARCHITECTURE

- MINISTRY OF SCIENCE AND TECHNOLOGY → ministerial umbrella
- DST → implementing department

MECHANISM / APPLICATION

- NQM → national capability mission
- T-HUBS → four vertical ecosystem anchors

READINESS / STATUS LIMIT

- RULE → department, mission and hub are distinct
- MINISTRY OF SCIENCE AND TECHNOLOGY → ministerial umbrella

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → department, mission and hub are distinct.

09

STAGE 09 FOUR T-HUB MAP

FOUR T-HUB MAP · IISC BENGALURU · QUANTUM COMPUTING · IIT MADRAS WITH C-DOT · QUANTUM COMMUNICATION · IIT BOMBAY · QUANTUM SENSING AND METROLOGY · IIT DELHI

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
IISc BENGALURU → Quantum Computing	IIT BOMBAY → Quantum Sensing and Metrology	TRAP → hub creation is not operational capability
IIT MADRAS WITH C-DOT → Quantum Communication	IIT DELHI → Quantum Materials and Devices	MUST REMEMBER: Quantum technology covers computing, communication and sensing: qubits...

SYSTEM / DEVICE / INSTITUTION

- IISc BENGALURU → Quantum Computing
- IIT MADRAS WITH C-DOT → Quantum Communication

DECISIVE TECHNICAL DISTINCTION

- IIT BOMBAY → Quantum Sensing and Metrology
- IIT DELHI → Quantum Materials and Devices

STATUS / UNIT / EVIDENCE LIMIT

- TRAP → hub creation is not operational capability
- MUST REMEMBER: Quantum technology covers computing, communication and sensing: qubits...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → hub creation is not operational capability.

10

STAGE 10 READINESS LADDER

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
IISc BENGALURU → Quantum Computing	IIT BOMBAY → Quantum Sensing and Metrology	TRAP → hub creation is not operational capability
IIT MADRAS WITH C-DOT → Quantum Communication	IIT DELHI → Quantum Materials and Devices	MUST REMEMBER: Quantum technology covers computing, communication and sensing: qubits...

SYSTEM / DEVICE / INSTITUTION

- IISc BENGALURU → Quantum Computing
- IIT MADRAS WITH C-DOT → Quantum Communication

DECISIVE TECHNICAL DISTINCTION

- IIT BOMBAY → Quantum Sensing and Metrology
- IIT DELHI → Quantum Materials and Devices

STATUS / UNIT / EVIDENCE LIMIT

- TRAP → hub creation is not operational capability
- MUST REMEMBER: Quantum technology covers computing, communication and sensing: qubits...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → hub creation is not operational capability.

10

STAGE 10 READINESS LADDER

READINESS LADDER · CABINET APPROVAL · MISSION AUTHORITY AND OUTLAY · INTENDED FUTURE CAPABILITY · RESEARCH PROTOTYPE · BOUNDED DEMONSTRATION · VALIDATED SYSTEM · REPEATABLE PERFORMANCE EVIDENCE

CABINET APPROVAL → mission authority and outlay

TARGET → intended future capability → RESEARCH PROTOTYPE → bounded demonstration → VALIDATED SYSTEM → repeatable performance evidence

STARTING CONDITION / INPUT

- CABINET APPROVAL → mission authority and outlay
- TARGET → intended future capability

SCIENTIFIC OR ENGINEERING MECHANISM

- RESEARCH PROTOTYPE → bounded demonstration
- VALIDATED SYSTEM → repeatable performance evidence

OUTPUT / FAILURE BOUNDARY

- OPERATIONAL DEPLOYMENT → field use with sustainment
- CLOSE DISTINCTION: Qubit is not a faster classical bit, gate count is not useful...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Qubit is not a faster classical bit, gate count is not useful.

11

STAGE 11 QUANTUM CLAIM FIREWALL

QUANTUM CLAIM FIREWALL · QUBIT, SUPERPOSITION, ENTANGLEMENT AND DECOHERENCE · COMPUTING, COMMUNICATION, SENSING OR MATERIALS · PHYSICAL AND LOGICAL QUBITS; QKD AND PQC · APPROVAL, TARGET, PROTOTYPE, VALIDATION AND OPERATION · OUTLAY, QUBITS, DISTANCE AND STARTUP CLAIMS OFFICIALLY · EVIDENCE LIMIT · IDENTIFY PLATFORM, QUBIT/MEASUREMENT

DEFINE → qubit, superposition, entanglement and decoherence → CLASSIFY → computing, communication, sensing or materials → SEPARATE → physical and logical qubits; QKD and PQC → LABEL → approval, target, prototype, validation and operation → VERIFY → outlay, qubits, distance and startup claims officially

SYSTEM / DEVICE / INSTITUTION

- DEFINE → qubit, superposition, entanglement and decoherence
- CLASSIFY → computing, communication, sensing or materials

DECISIVE TECHNICAL DISTINCTION

- SEPARATE → physical and logical qubits; QKD and PQC
- LABEL → approval, target, prototype, validation and operation

STATUS / UNIT / EVIDENCE LIMIT

- VERIFY → outlay, qubits, distance and startup claims officially
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Identify platform, qubit/measurement...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Identify platform, qubit/measurement...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Identify platform, qubit/measurement.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · QUANTUM CAPABILITY IS PATH-DEPENDENT · NATIONS THAT BUILD TALENT, FABRICATION KNOW-HOW AND STANDARDS EARLY · THE MOST BINDING CONSTRAINTS ARE OFTEN ENGINEERING AND ECOSYSTEM · A MISSION APPROACH IS JUSTIFIED BECAUSE FRAGMENTED PROJECTS ALONE · NATIONAL MISSION FRAMEWORK. · T-HUB AT IISC BENGALURU · QUANTUM COMPUTING.

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- CAUTION: Quantum capability is path-dependent: nations that build talent, fabrication know-how and standards early can gain long-term advantages.
- CAUTION: The most binding constraints are often engineering and ecosystem constraints rather than pure theory.
- CAUTION: A mission approach is justified because fragmented projects alone may not create platform-level capability.
- NQM: national mission framework.

READINESS, UNIT OR STATUS LIMIT

- T-Hub at IISc Bengaluru: Quantum Computing.
- T-Hub at IIT Madras with C-DOT: Quantum Communication.
- T-Hub at IIT Bombay: Quantum Sensing & Metrology.
- T-Hub at IIT Delhi: Quantum Materials & Devices.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- PIB parliamentary answer: says the T-Hubs comprise 14 Technical Groups across 17 States and 2 UTs.
- Quantum communication is strategically relevant for secure networks and national-security communication contexts.
- Quantum sensing/metrology can matter for navigation, clocks, precision instrumentation and advanced measurement ecosystems.
- Quantum materials/devices are foundational because hardware dependence can otherwise cripple mission ambition.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels